

How Can the Integrated Routing and Scheduling Enhance Optimality Bounds of Time-Sensitive Transmission

Zhenrui Cao, Tie Qiu, Qingyong Deng, Haolin Liu, and Xiaobo Zhou

ABSTRACT

Time-sensitive networking (TSN) ensures reliable real-time transmission services for diverse applications in the Industrial Internet of Things through strictly deterministic time-triggered (TT) flows. The end-to-end transmission performance of TT flows is jointly influenced by routing choices and bandwidth scheduling, making their optimization essential for improving the overall service quality of TSN networks. Existing methods address bandwidth scheduling only after determining the route, leading to a disjointed approach to routing and scheduling decisions. This separation may introduce potential transmission conflicts, such as routes with insufficient available bandwidth being selected prematurely, resulting in the failure to allocate the required bandwidth resources for deterministic flow transmission. To maximize the transmissibility of TT flows, we propose MAIRS, a multi-agent deep reinforcement learning (MADRL)-based Integrated Routing and Scheduling method. First, leveraging network softwareization, we design a synchronized decision-making framework for integrated routing and scheduling to enhance the optimality bounds of TT flow transmission. Next, we introduce collaborative MADRL to develop comprehensive end-to-end transmission plans for TT flows, effectively addressing the intricate integrated routing and scheduling problem characterized by a vast solution space. Experiments demonstrate that MAIRS significantly outperforms existing algorithms in improving the transmission performance of TT flows.

INTRODUCTION

Compared to traditional best-effort Ethernet, time-sensitive networking (TSN) ensures deterministic, reliable, and low-latency transmission services, playing an irreplaceable role in supporting real-time applications within the Industrial Internet of Things (IIoT) [1]. In TSN networks, time-triggered (TT) flows are prioritized, with their delay and jitter conforming to stringent constraints [2]. TT flows ensure strict determinism and serve as the primary conduit for real-time data transmission [3]. Consequently, closely meeting the determinism requirements of TT flows is key to

enhancing the practical effectiveness of industrial TSN networks in serving industrial applications.

In TSN networks, the routing and scheduling schemes determine the end-to-end transmission performance of TT flows [4]. To minimize bandwidth occupancy conflicts between TT flows, routes must be dynamically determined based on the transmission demands and real-time network conditions [5]. Scheduling involves allocating dedicated transmission time slots for each TT flow, aligning with the TSN traffic shaping mechanisms to ensure deterministic end-to-end transmission [3]. Considering the flexible network management enabled by softwareization technology, the development of software-defined routing and scheduling optimization can facilitate the long-term effective utilization of bandwidth resources within TSN networks, thereby maximizing the successful transmission of TT flows [2].

In TSN networks, routing entails managing network resources at the link level, while scheduling pertains to bandwidth allocation at a finer granularity through slots [2]. The inconsistency between these two management levels inherently hinders existing methods from realizing true joint optimization of routing and scheduling. Specifically, these methods depend on coarse link-level metrics, such as bandwidth utilization, to initially determine the route for a TT flow [4, 5]. Subsequently, within the optional time slot range specified by the TT flow on the selected route, they attempt to allocate a set of available slots for transmission. However, under this separated and sequential decision-making mode, routing decisions often prioritize general link-level metrics while neglecting the meticulous management of slot resources. This can result in a situation where, despite selecting an overall smooth route, all slots within the TT flow's optional range are occupied, causing subsequent slot scheduling to fail and leading to nondeterministic transmission of the TT flow. This decision-making approach essentially fragments the overall optimization space of end-to-end transmission, potentially causing conflicts between routing and scheduling, thereby failing to maximize the transmissibility of TT flows.

Conversely, integrating routing and scheduling

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at the slot granularity, by synchronously determining both routing and scheduling plans, can effectively eliminate conflicts that may arise in the asynchronous decision-making mode, thereby unlocking the full optimization potential for TT flow transmission [2]. However, this integration also poses significant challenges, as it results in an increased variable scale and deepens constraint dependencies, creating a vast solution space that complicates the search for global optimality. While recent studies on optimizing TSN transmission demonstrate that deep reinforcement learning (DRL) can outpace knowledge-dependent heuristic methods through autonomous exploration and learning, a single DRL agent encounters difficulties in integrated routing and scheduling optimization due to its limited capacity to navigate the expansive, high-dimensional nonlinear solution space [6, 7]. In contrast, the collaborative multi-agent deep reinforcement learning (MADRL) framework centered on a shared global optimization objective decouples end-to-end transmission optimization into multiple subproblems, thereby circumventing the curse of dimensionality encountered by single-agent DRL and enhancing decision-making through agent collaboration [8, 9].

Building on these insights, we propose MAIRS, a MADRL-based Integrated Routing and Scheduling method aimed at maximizing the transmissibility of TT flows. The main contributions of this article are as follows:

- We design a softwarized transmission management framework that integrates routing and scheduling for TT flows, enhancing the optimality bounds of transmission optimization by eliminating potential conflicts between routing and scheduling schemes.
- We introduce collaborative MADRL to decompose the end-to-end transmission optimization problem, characterized by a vast solution space, into multiple subproblems addressed by individual agents. Furthermore, we ensure that comprehensive end-to-end transmission plans are generated through efficient agent collaboration.
- We conduct experiments to demonstrate that MAIRS significantly improves the overall transmission performance of TT flows in industrial TSN networks.

In the following sections, we first review flow management in TSN, focusing on routing and scheduling for TT flows. We then detail the proposed MAIRS, and present the algorithm evaluation. The final section concludes the article.

BACKGROUND

FLOW MANAGEMENT IN TSN NETWORKS

TSN, developed by the IEEE 802.1 working group, comprises a series of standards that provide deterministic delay bounds for real-time applications [6]. The centralized management architecture proposed by IEEE 802.1Qcc enables flexible configuration and dynamic management of industrial TSN networks through software-defined approaches, such as the OpenFlow protocol [2, 10]. Figure 1 depicts the typical architecture of software-defined management for industrial TSN networks. Differentiated devices are connected to the TSN network via full-duplex links through TSN

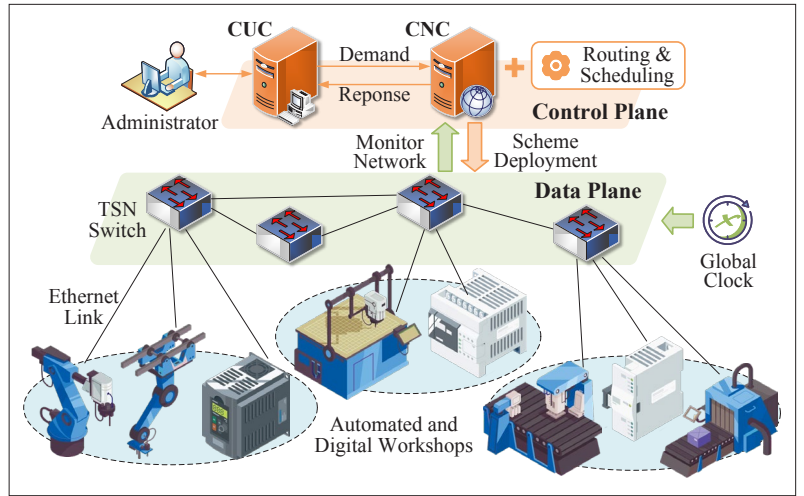


FIGURE 1. The architecture of software-defined management for industrial TSN networks.

switches. Switches in the data plane, guided by a global clock for time synchronization, are responsible for packet forwarding, with their behavior governed by the control plane. The centralized user configuration (CUC) in the control plane offers an interface for administrators and is tasked with parsing transmission demands to the centralized network configuration (CNC) [10]. Based on its global view of the topology, switches, and flows in the data plane, CNC utilizes routing and scheduling algorithms to generate end-to-end transmission plans in response to transmission demands. Network softwarization enables the dynamic deployment of transmission plans on the data plane to update the forwarding rules of TSN switches. Through flexible traffic management and dynamic network control, softwarization streamlines the configurations required to ensure determinism in industrial TSN networks.

In TSN, flows are classified into three service levels with decreasing priority: TT, audio-video bridging (AVB), and best-effort (BE). TT flows are guaranteed strict determinism to support real-time industrial applications with stringent delay requirements [2]. To enable effective management of flows with different priorities, the egress ports of TSN switches are outfitted with eight queues, each designated for a specific flow type. TSN also defines various traffic shaping mechanisms, such as the time-aware shaper (TAS), to ensure that critical TT flows experience controlled delays without facing unexpected bandwidth preemption from lower-priority flows [3]. These mechanisms enable bandwidth allocation in different modes to regulate traffic forwarding behavior. Among them, TAS, as defined by the IEEE 802.1Qbv standard, is the most widely used shaping mechanism [11]. Under the condition of network-wide time synchronization, TAS manages the opening and closing of queue gates according to the gate control lists (GCL), facilitating flexible and precise management of TT flows across the entire network [2]. The egress port model of a TSN switch equipped with TAS is shown in Fig. 2. TSN switches typically permit custom matching of flow types to queues; in this example, TT flows occupy queue Q_0 , while the remaining seven queues are utilized by AVB and BE flows. Upon arrival at the switch, packets undergo processing through the

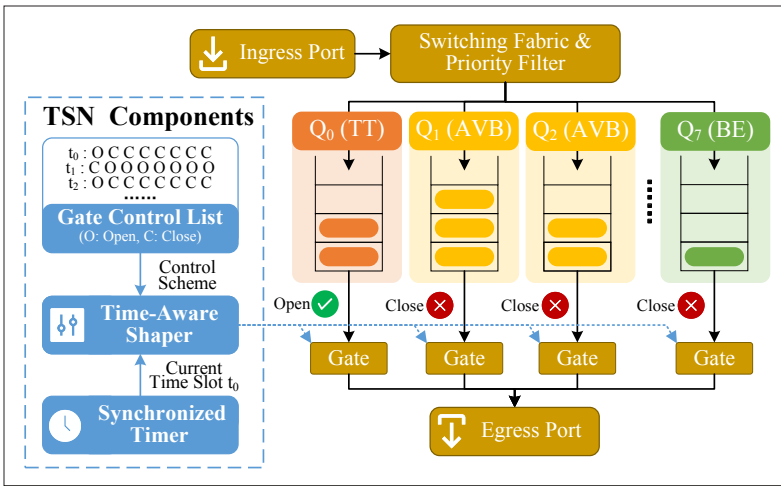


FIGURE 2. Egress port model of a TSN switch deployed with TAS.

switching fabric and priority filter, then classified into their corresponding waiting queues. When the network-wide synchronized timer indicates the current time slot as t_0 , TAS controls the opening and closing of queue gates according to the corresponding entry in the GCL, allowing only TT flows in Q_0 to transmit while blocking lower-priority flows. TAS separates the transmission of TT flows and lower-priority flows into distinct slots, thereby avoiding unexpected bandwidth preemption by lower-priority flows and ensuring the determinism of TT flow transmission.

ROUTING AND SCHEDULING FOR TT FLOWS

In recent years, numerous studies have emerged focusing on optimizing flow transmission in TSN networks. Among these, DRL-based scheduling algorithms have achieved state-of-the-art performance. Yang *et al.*, by refining a multi-queue model to enhance deterministic guarantees for bursty flows, utilized a two-dimensional feature mapping of slots and queues to assist DRL in maximizing the number of schedulable flows [3]. Wang *et al.* employed a DRL-enhanced ant colony algorithm to optimize end-to-end transmission in the integrated TSN and 5G network [6]. Yang *et al.* have further applied DRL to optimize TSN-based federated learning [12]. These algorithms focus solely on optimizing slot scheduling for flow transmission, neglecting the crucial impact of routing on transmission performance. Conversely, Nayak *et al.* and Zhou *et al.* proposed optimization models that combine routing and scheduling, and employed exact algorithms to determine end-to-end transmission schemes [2, 13]. However, these methods are impractical due to their high computational cost. Li *et al.* and Zhang *et al.* developed heuristic rules to initially determine the route for a TT flow, subsequently allocating slots along the selected route [5, 14]. He *et al.* introduced a scalable route feature extractor and employed DRL to select routes for the TT flows, followed by heuristic-based slot scheduling [4]. However, these methods are constrained by the inconsistency in bandwidth resource management levels between routing and scheduling. Separating the decision-making for routing and scheduling introduces potential conflicts between the two, hindering the optimization of TT flow transmis-

sion. In contrast, integrating routing and scheduling by aligning them at the slot granularity can enhance the upper bounds of optimality for end-to-end transmission optimization.

We then use the simple example shown in Fig. 3 to further illustrate the necessity of integrating routing and scheduling decisions. To ensure the end-to-end determinism of TT flow transmission, conflict-free time slots must be allocated on all links along the route. The TT flow entering switch S_1 before time slot t_0 must reach switch S_5 before slot t_5 . The method proposed by Li *et al.* initially heuristically selects S_1 - S_2 - S_4 - S_5 with lower overall utilization as the routing plan [5]. However, in the subsequent slot scheduling phase, it is found that no conflict-free time slot allocation on this route satisfies the TT flow's delay requirement, thus impeding successful transmission. Integrating routing and scheduling entails aligning them at the same slot level, enabling the selection of alternative, conflict-free routing and scheduling plans through the perception and analysis of fine-grained slot occupancy states. Although S_1 - S_3 - S_4 - S_5 is overall more congested, it contains a slot scheduling scheme that meets the flow transmission requirements. Integrating routing and scheduling prevents the disconnection between routing and scheduling plans, ensuring the maximum optimization potential for end-to-end flow transmission based on a comprehensive understanding of the network's fine-grained state.

MAIRS: MADRL-BASED INTEGRATED ROUTING AND SCHEDULING

END-TO-END TRANSMISSION MANAGEMENT FRAMEWORK

Driven by the theoretical advancements offered by integrated routing and scheduling for TSN transmission optimization, we design an end-to-end transmission management framework based on MADRL. We will further elucidate this framework with Fig. 4. Leveraging software-defined management of TSN networks, we incorporate a multi-agent cluster into the CNC, establishing it as the decision-making core for generating integrated routing and scheduling plans for TT flow transmission. Each agent in the cluster is logically mapped to a specific TSN switch in the data plane. During the decision-making phase, individual agents will dictate the local transmission behavior of TT flows, including routing allocation and slot occupancy, at their corresponding switches. Consequently, the logical interactions between agents also map to the connectivity of the switches, reflecting the direct collaboration relationships among the agents.

The transmission management process focused on integrated routing and scheduling is divided into five steps, as shown in Fig. 4. Whenever a new transmission demand arises, the CUC first parses the flow information to extract key attributes, including the origin and destination of transmission. The parsed results are then transmitted to the agent cluster, where the agent assigned to the origin switch generates the local integrated routing and scheduling scheme for the TT flow at the origin hop. This local scheme specifies which time slot the flow should be transmitted in and to which adjacent switch. In the subsequent

hop-by-hop decision-making process, the corresponding agent at the next-hop switch receives and processes this transmission demand, continuing to generate the local integrated routing and scheduling scheme for the flow. This process continues until a complete transmission scheme that fulfills the transmission demand is found, or if any agent makes an invalid decision, causing the decision process to terminate. Once hop-by-hop decision-making is finished, the generated aggregated scheme is transferred to the CNC for further evaluation. If the aggregated scheme is assessed as feasible, the CNC deploys it to the data plane, allowing the flow to be reliably transmitted according to the planned routing and scheduling scheme.

Under this management framework, agents collaboratively generate integrated routing and scheduling plans for end-to-end transmission. Agents are responsible for local decisions at each hop to ensure that the routing and scheduling schemes for TT flows are well coordinated. The synchronized decision-making pattern of routing and scheduling fully reveals the interdependencies between routing choices and scheduling plans. This coordination mechanism effectively resolves potential conflicts between routing and scheduling, thereby enhancing the optimality bounds of TT flow transmission.

However, the interdependence of routing and scheduling at the slot level leads to an exponential increase in the number of variables, thereby intensifying the complexity of combinatorial optimization and making the search for global optimality more challenging. To address this, the MADRL system is introduced and designed as a collaborative decision-making mode, as shown in Fig. 4, leveraging the nature of TT flows being forwarded hop-by-hop across TSN switches. The MADRL-based transmission management framework decouples the end-to-end transmission optimization which has a vast solution space, into multiple subproblems that can be solved independently by the agents. This approach mitigates the curse of dimensionality inherent in complex optimization issues, enabling flexible and efficient flow management.

DESIGN OF COLLABORATIVE MADRL

The hop-by-hop collaboration among the agents aligns with the characteristics of a multi-agent Markov decision process (MAMDP), as each agent's local decision is influenced not only by the local slot-level network states but also by the decisions made by preceding agents. The hop-by-hop collaboration will form an ordered chain of actions, with each agent's local action guided by the preceding agents, creating a coherent overall decision-making process. We expect each agent to receive effective training, enabling efficient collaboration to maximize the number of transmittable TT flows. To achieve this goal, we will now present the specific design of the agents.

Observation Design: Each agent's input consists of two parts: global observation and local observation. This global observation remains consistent across all agents participating in the decision-making for the current TT flow transmission, primarily including key information such as the attributes of the TT flow, the network topology, and the bandwidth utilization across links. The

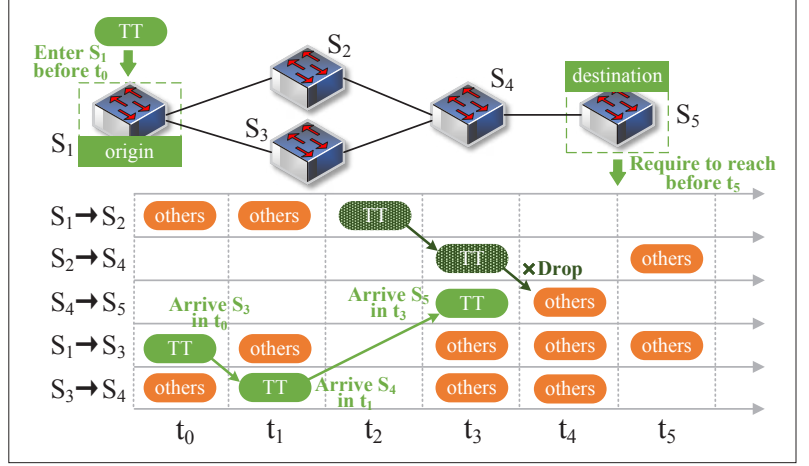


FIGURE 3. A decision-making example of routing and scheduling.

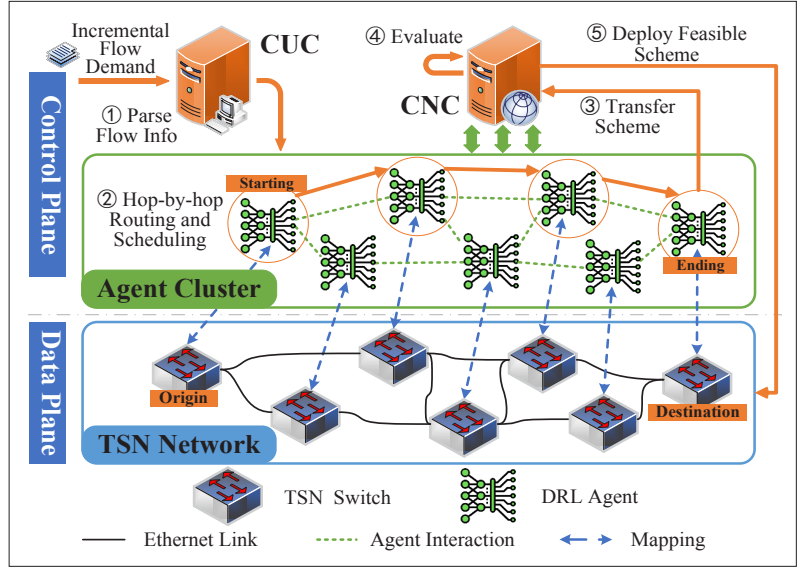


FIGURE 4. The management framework of TT flow transmission.

flow attributes cover essential elements like the origin, destination, transmission period, and the size of the data being transmitted.

The local observation focuses on the unique context of each agent. It encompasses the action chain of the preceding agents and the detailed bandwidth utilization of all optional time slots for local hop transmission. By analyzing the action chain, which includes accumulated delays and the identities of the agents involved in the transmission, the current agent can make informed decisions regarding integrated routing allocation and slot occupancy. Given that the action chain functions as the communication channel between agents, it is crucial for the agents to fully comprehend it to ensure effective collaboration among them. The range of optional slots is constrained by the frame transmission period of the TT flow. Specifically, the latest optional time slot of a frame must not coincide with the transmission period of the subsequent frame to maintain the periodicity of TT flow transmission.

Action Design: The decision-making process at each agent is unique, as each corresponds to a switch with a specific connection status, leading to different available action options for each agent.

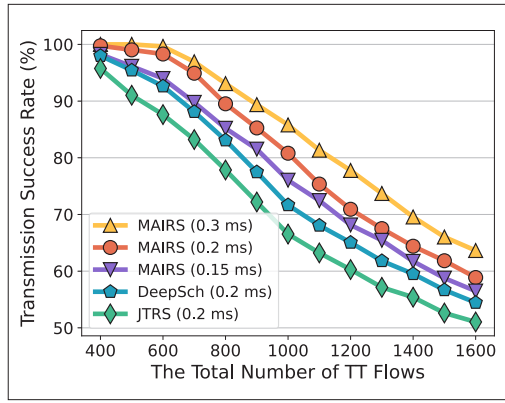


FIGURE 5. Comparison regarding transmission success rate under varying total number of flows.

Specifically, the number of possible actions for an agent is calculated as the product of the number of neighboring agents and the maximum range of optional slots for all TT flows. From these options, the agent selects the most suitable action, forming a local cohesive routing and scheduling plan.

However, not every action option is feasible. Some choices may result in error situations where the deterministic transmission of the TT flow cannot be guaranteed, thereby forcing the current TT flow's hop-by-hop decision-making process to terminate prematurely. These error situations include: the formation of routing loops, which cause packets to circulate within the network; selecting integrated routing and scheduling schemes that fail to ensure conflict-free bandwidth usage; and resulting in excessive delays that hinder transmission from completing within the required time. Each error situation negatively impacts the reliability and efficiency of TT flow transmission. Therefore, agents must carefully select their actions to avoid these potential pitfalls.

Reward Design: After each round of hop-by-hop decision-making, we provide unified feedback to all involved agents. Specifically, when the end-to-end transmission plan successfully directs the TT flow to its intended destination, each agent participating in the decision-making process receives positive feedback of $x = 1$. Conversely, if any agent triggers an error situation, all agents involved receive a penalty of $x = -1$. Additionally, we also assess the network's status by measuring the bandwidth utilization of the most heavily loaded time slot. This utilization rate is denoted as $0 \leq y \leq 1$. Load-balanced networks can typically accommodate a greater number of flows; therefore, we also incorporate y into the agent feedback [3]. Ultimately, the reward for agents is defined as $x - \alpha \cdot y$, where $0 \leq \alpha \leq 1$ is a scaling factor.

The design of the reward function, which is calculated as the transmission success indicator minus the maximum slot utilization rate, serves several important purposes. Firstly, it incentivizes agents to consider the broader implications of their actions on overall transmission, fostering more thoughtful decision-making, as agents must weigh the benefits of successful transmissions against the potential for failures. Secondly, by penalizing actions that lead to high bandwidth utilization, the reward function promotes load balancing across the network. This is vital in avoiding bottlenecks that could impair

the transmission of the TT flows over time. Ultimately, this design fosters a collaborative environment where agents are motivated to achieve the common objective of the goal of maximizing the number of transmissible TT flows.

Training Framework: We train multiple agents collaboratively using multi-agent proximal policy optimization (MAPPO) based on the actor-critic architecture [15]. During the training process, the critic network of each agent utilizes not only its own local observations but also additional local observations from other agents. This information sharing allows the critic network to develop a more comprehensive understanding of the overall state for the current flow's transmission, leading to more precise estimates of the critic values. This approach will further implicitly strengthen the actor network's decision-making capability to make more informed decisions, promoting better cooperation among agents. Additionally, MAPPO improves the stability of agent policy updates through a clipping mechanism. Specifically, MAPPO restricts the extent of policy changes during each update to prevent drastic fluctuations in the policy during training.

Introduction of Action Mask: In the early stages of training, agents with randomly initialized policy parameters often select local actions that result in error situations, leading to premature termination of the current decision-making round [8]. These frequent errors can hinder the agents' ability to effectively explore efficient transmission strategies. To mitigate this issue, we introduce action masks to obscure invalid local actions. Specifically, agents will set the sample probabilities of invalid local actions to zero based on the provided action masks, thereby preventing their selection [8]. This design helps avoid errors and enables collaborating agents to learn more effective integrated routing and scheduling strategies.

EVALUATION

To comprehensively evaluate the advantages of MAIRS, we select DeepSch and JTRS as comparison algorithms. DeepSch designs a scalable route feature extractor, using DRL to determine the most suitable route from multiple available shortest paths for the TT flow [4]. After determining the route, it follows the "earliest-valid-slot-first" principle to allocate time slots along the selected route. JTRS, on the other hand, heuristically determines routes for TT flows by considering link load and hop count [5]. Subsequently, it prioritizes time slots that effectively reduce end-to-end delay for scheduling. Both DeepSch and JTRS consider the impact of routing and scheduling on TT flow transmission, but unlike MAIRS, they employ an asynchronous combination of routing and scheduling. Comparing these two algorithms highlights how MAIRS enhances the optimality bounds of time-sensitive transmission through the integration of routing and scheduling.

EXPERIMENT SETUP

In the experiments, we select the Orion Crew Exploration Vehicle (CEV) network, widely employed in previous research, as the benchmark switch topology for evaluation [3]. This network integrates star, tree, and ring structures to ensure comprehensive and representative testing. Due to the lack of publicly available industrial application

datasets, the following network configurations are determined by referencing typical setups from previous work [3, 4, 6]. The link bandwidth in the network is uniformly set to 100 Mb/s. TT flows are randomly distributed across the network, with data sizes ranging from 60 to 1500 bytes and transmission periods randomly chosen from the set {3.6, 4.8, 6.0} ms. The duration of time slots is selected from 0.3, 0.2, and 0.15 ms, enabling us to conduct a sensitivity analysis of transmission optimization in relation to different bandwidth management granularities. Specifically, we configure five different combinations of algorithms and slot durations. MAIRS is further denoted as MAIRS (0.3 ms), MAIRS (0.2 ms), and MAIRS (0.15 ms), corresponding to different slot duration conditions, respectively. DeepSch and JTRS will be compared with MAIRS at the slot duration of 0.2 ms, referred to as DeepSch (0.2 ms) and JTRS (0.2 ms), respectively.

In MAIRS, the value of the scaling factor α is determined to be 0.25 through grid search. During training, the agents undergo 8,000 iterations to achieve policy convergence. The learning rate is set to 0.003, with a batch size of 32 samples. To ensure reliability, DeepSch and JTRS follow their respective recommended parameter settings [4, 5]. The experiments are conducted in 30 independent test environments to evaluate the performance of different algorithms.

COMPARISON OF TT FLOW TRANSMISSION

We assess the algorithm performance with varying numbers of TT flows to be transmitted. Figure 5 illustrates the success rates of transmission, while Fig. 6 presents the utilization of bandwidth resources during the experiments. As the number of TT flows increases and congestion in the network intensifies, the transmission success rate gradually decreases, while bandwidth resources are utilized more effectively. Thanks to the integrated optimizations of routing and scheduling by the collaborative multi-agent system for end-to-end transmissions, MAIRS demonstrates superior transmission performance compared to DeepSch and JTRS when the slot duration is set to 0.2 ms. Specifically, as the total number of flows increases from 400 to 1000, the transmission success rate of MAIRS improves over the suboptimal algorithm DeepSch by 1.80 percent to 12.74 percent. However, as the flow count rises from 1000 to 1600, this improvement gradually decreases to 8.14 percent. Correspondingly, MAIRS also achieves its maximum improvement in bandwidth utilization of 6.79 percent compared to DeepSch when the flow count is 1000. As the flow count increases from 400 to 1000, the heightened bandwidth contention among flows elevates the complexity of the optimization problem and amplifies the efficient optimization capability of MAIRS, thus gradually expanding its advantage during this phase. However, when the flow count exceeds 1000, MAIRS remains superior to other algorithms, but the congestion effects caused by the excessive flows limit the potential for further optimization, resulting in a slight reduction in its margin of advantage. JTRS performs worse than DeepSch, primarily due to its heavy reliance on expert-configured heuristics, which leads to poorer adaptability to the problems.

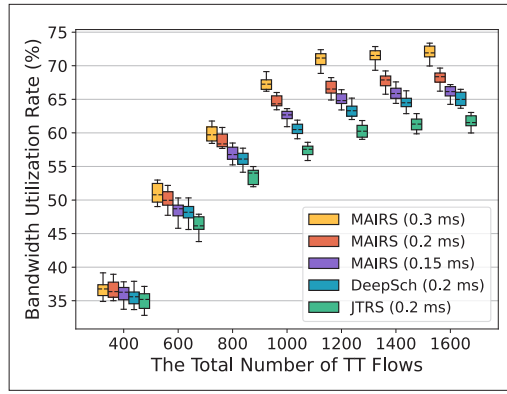


FIGURE 6. Comparison regarding bandwidth utilization rate under varying total number of flows.

Experimental results presented in Figs. 5 and 6 also reveal that MAIRS exhibits varying transmission performance under different slot duration conditions. With increments of 200 flows from 400 to 1600, MAIRS (0.3 ms) improves the transmission success rate by 0.25 percent, 1.36 percent, 3.98 percent, 6.18 percent, 9.65 percent, 8.01 percent, and 8.09 percent compared to MAIRS (0.2 ms). In comparison to MAIRS (0.15 ms), the success rates increase by 1.79 percent, 6.02 percent, 9.25 percent, 12.83 percent, 14.10 percent, 12.68 percent, and 12.66 percent, respectively. Consistently, longer slot durations result in higher bandwidth utilization rates for MAIRS. In contrast, shorter slot durations impede transmission optimization by fragmenting and dispersing bandwidth resources, thereby obstructing efficient utilization.

CONCLUSION AND FUTURE WORK

In this article, we propose MAIRS, an integrated routing and scheduling method based on collaborative MADRL to optimize the transmissibility of TT flows. We first introduce a softwareized flow management framework designed to enhance the optimality bounds of end-to-end transmission optimization for TT flows while addressing potential conflicts between routing and scheduling schemes. Based on this framework, we introduce collaborative MADRL to decompose the end-to-end transmission optimization problem, characterized by an extensive solution space, into multiple subproblems. Experimental results show that, compared to other advanced approaches, MAIRS effectively increases the number of transmissible TT flows. In future work, we will continue to enhance industrial TSN networks to meet the diverse demands of various dynamic industrial applications, advancing the adoption and development of TSN technology in practical industrial environments.

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